

A 3×3 Counterexample to Convexity

This example isolates the source of nonconvexity in the physical CML data-fidelity term. For one link,

$$\hat{A}(R) = \sum_{p \in \ell} d_p k R_p^\alpha, \quad J_{\text{atten}}(R) = \frac{(\hat{A}(R) - A_{\text{obs}})^2}{L}. \quad (1)$$

This is the one-link normalization used by the normalized solver.

1 Toy Link

The example has the following parameters:

- a 3×3 pixel grid;
- one horizontal link crossing the three pixels in the middle row;
- a crossed length of 0.125 km in each pixel, giving a total link length of $L = 0.375$ km;
- a frequency of 38.003 GHz with horizontal polarization;
- the ITU-R P.838-3 coefficients $k = 0.400171594750$ and $\alpha = 0.881535333268$.

Consider two strictly positive rain fields, measured in mm/h, that are left-right reflections of each other across the vertical centerline of the grid. In $R^{(L)}$, the heavy-rain value is in the middle-left pixel. In $R^{(R)}$, the same value is in the middle-right pixel:

$$R^{(L)} = \begin{bmatrix} 1 & 1 & 1 \\ 20 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad R^{(R)} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 20 \\ 1 & 1 & 1 \end{bmatrix}. \quad (2)$$

Either field could be chosen as the ground truth. If $R^{(L)}$ is the ground truth, it generates the observation below, but $R^{(R)}$ produces exactly the same observation. The same statement holds with the roles reversed. The single horizontal link therefore cannot distinguish these two rain fields. Because the link crosses the reflected pixels over equal distances, their predicted attenuations are equal:

$$\hat{A}(R^{(L)}) = \hat{A}(R^{(R)}) = A_{\text{obs}} = 0.801595407171 \text{ dB}. \quad (3)$$

It follows that

$$J_{\text{atten}}(R^{(L)}) = J_{\text{atten}}(R^{(R)}) = 0. \quad (4)$$

The arithmetic mean of the two fields is

$$R^{(M)} = \frac{R^{(L)} + R^{(R)}}{2} = \begin{bmatrix} 1 & 1 & 1 \\ 10.5 & 1 & 10.5 \\ 1 & 1 & 1 \end{bmatrix}. \quad (5)$$

2 Why the Power Law Is Strictly Concave

To see why the power law is strictly concave, define $g(r) = r^\alpha$ for $r > 0$. Its second derivative is

$$g''(r) = \alpha(\alpha - 1)r^{\alpha-2}. \quad (6)$$

Here, $\alpha > 0$, $\alpha - 1 < 0$, and $r^{\alpha-2} > 0$. Therefore, $g''(r) < 0$ for every $r > 0$, which means that g is strictly concave. In particular,

$$\left(\frac{20+1}{2}\right)^\alpha > \frac{20^\alpha + 1^\alpha}{2}. \quad (7)$$

In the arithmetic-mean field, the two reflected rain rates 20 and 1 become 10.5 and 10.5. The strict-concavity inequality shows that their combined attenuation contribution increases. Consequently,

$$\hat{A}(R^{(M)}) = 0.845082934781 \text{ dB} > A_{\text{obs}}, \quad (8)$$

and

$$J_{\text{atten}}(R^{(M)}) = 0.005043106820 > 0. \quad (9)$$

Therefore,

$$J_{\text{atten}}\left(\frac{R^{(L)} + R^{(R)}}{2}\right) > \frac{J_{\text{atten}}(R^{(L)}) + J_{\text{atten}}(R^{(R)})}{2} = 0. \quad (10)$$

This violates the defining inequality for a convex function. Therefore, the physical attenuation data term is nonconvex.